

Lecture 2:

What can I amplify?

ECE 340 – Electronic Circuits I Sec. 1

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09/08/2026

Material credit: Prof. Jennifer Volk



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and Computer Engineering
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Announcements

Homework 1 to be posted tonight. Due on next Tuesday 11:59 pm.

Tentative exam dates:

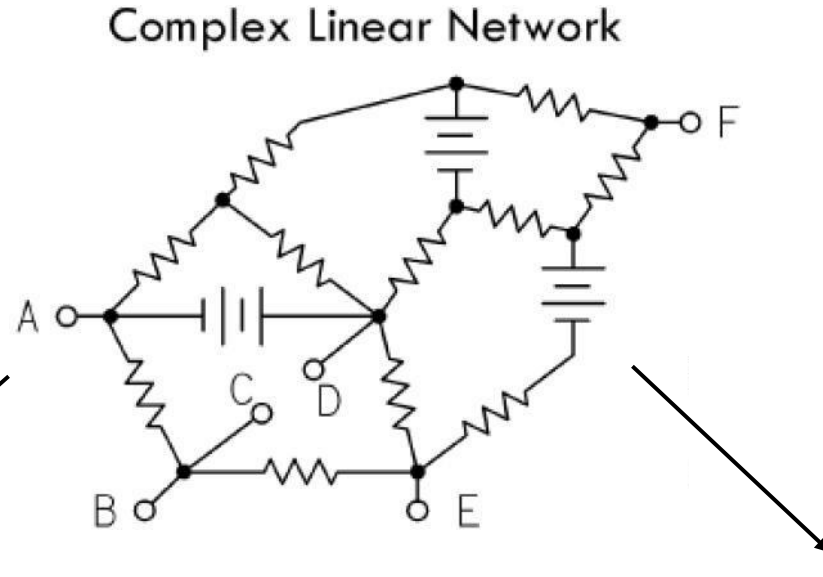
Midterm 1: 10/8/2026 during class time

Midterm 2: 11/12/2026 during class time

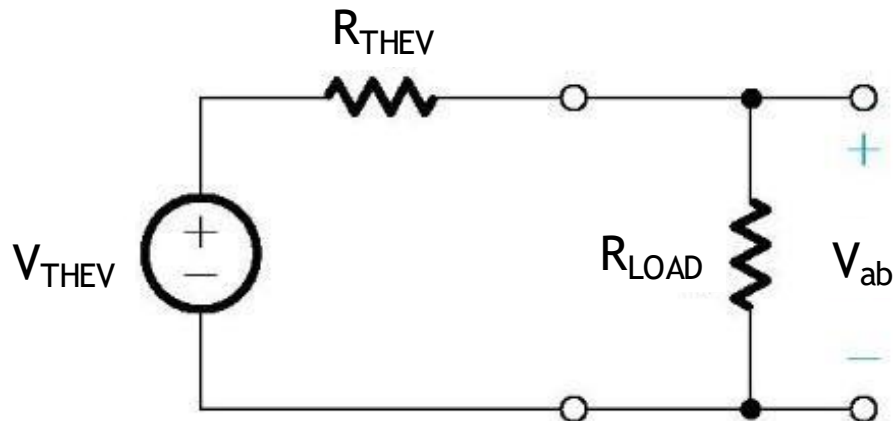


Last time...

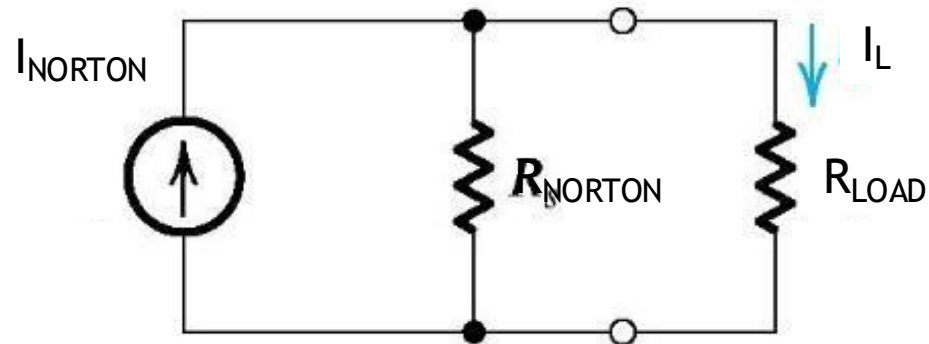
Some circuits can be too complex to analyze by hand



Thevenin's theorem: Any linear two-terminal dc network can be represented by a voltage source in series with a resistor.



Norton's theorem: Any linear two-terminal dc network can be represented by a current source in parallel with a resistor.

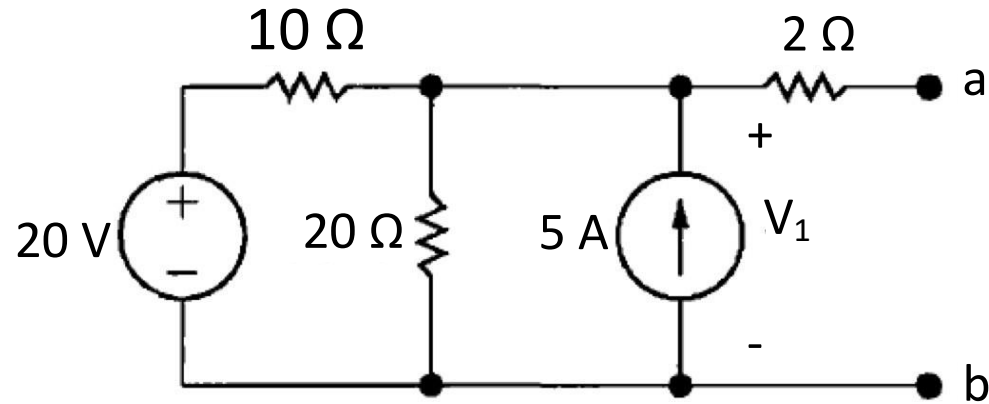


Example 4

Find Norton equivalent

Step 1: Find I_{NORTON} by replacing load with a short.

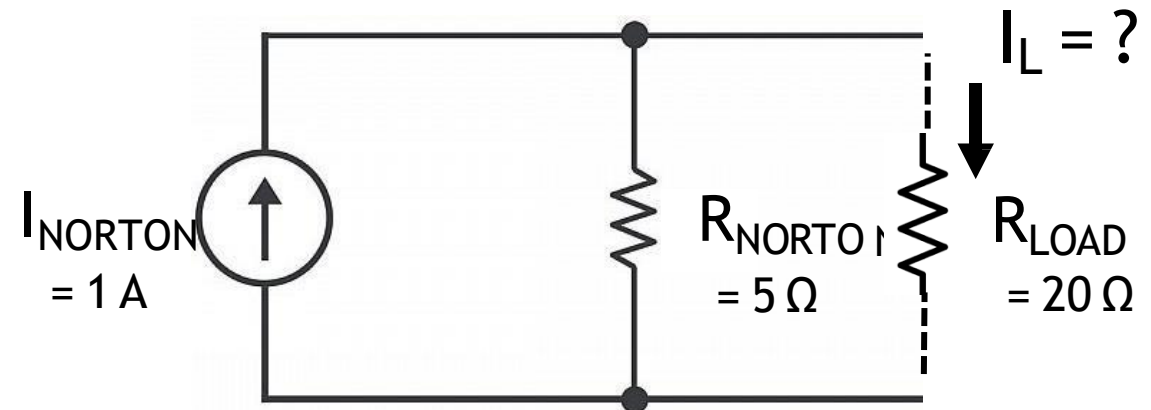
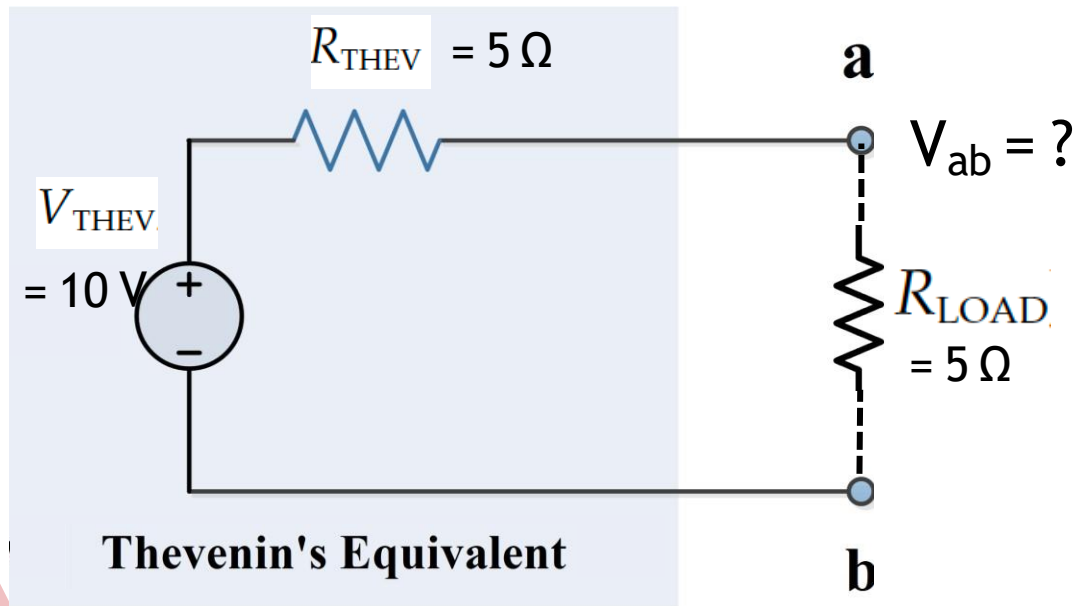
Step 2: Zero the sources and open R_{LOAD} to get R_{NORTON} .

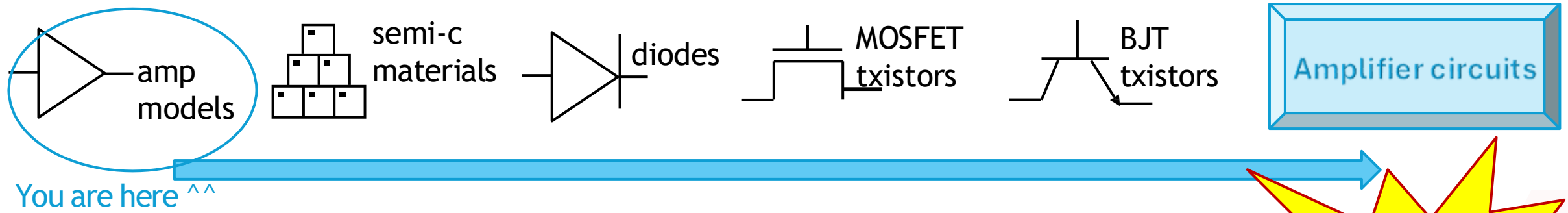


An important note

Although you know what your V_{THEV} and I_{NORTON} values are, the **DELIVERED Voltage/Current to R_{LOAD} can be DIFFERENT** depending on the ratio between $R_{\text{THEV}}/R_{\text{NORTON}}$ and R_{LOAD} .

Find V_{ab} and I_L in the following examples to see this in practice!



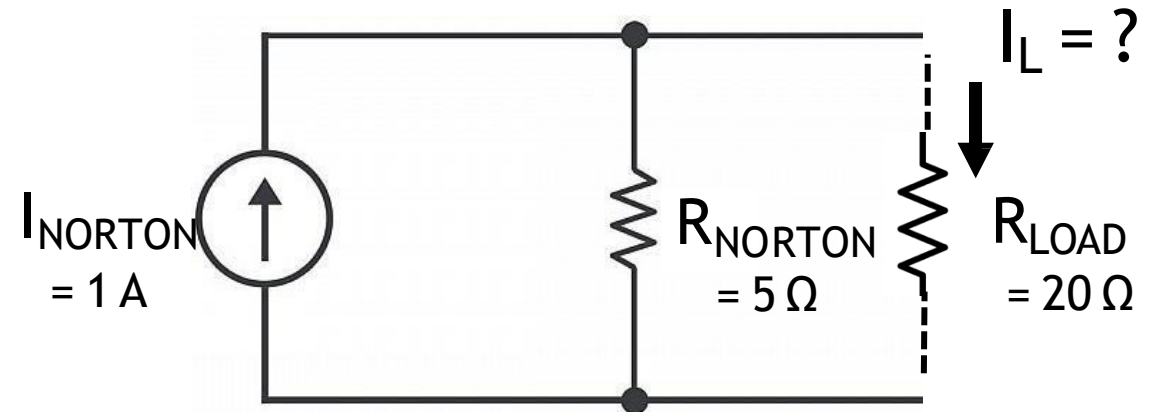
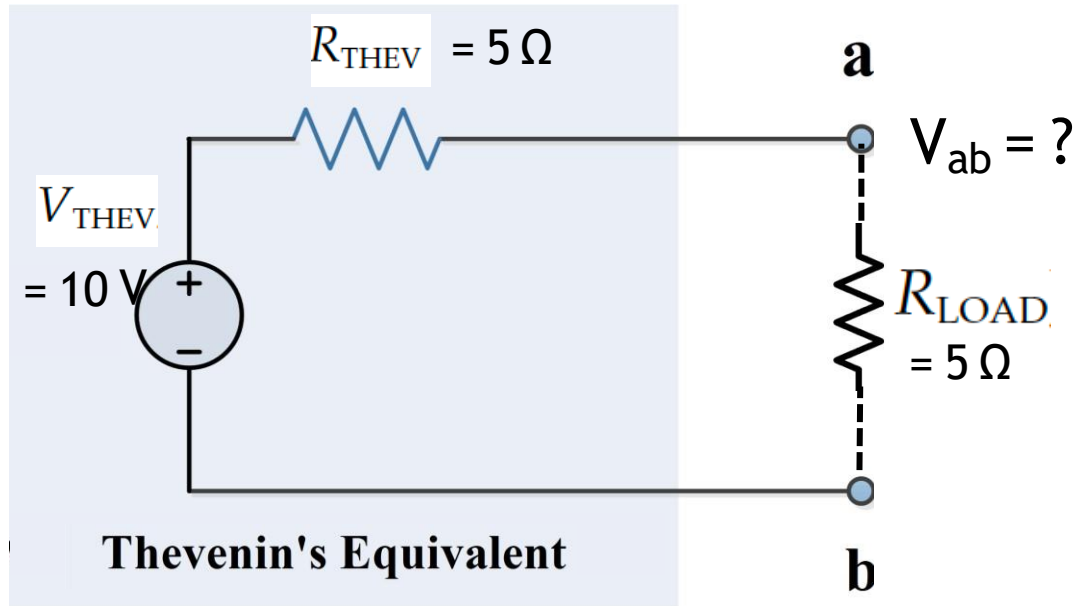


Today's Outline

1. Voltage C current dividers
2. Dependent sources C source absorption
3. Ideality of signal sources
4. Voltage/Current/Power gain

New! Class progress bar

Output voltage and current differs



Method 1:

$$I = 10\text{ V} / (5\ \Omega + 5\ \Omega) = 1\text{ A}$$
$$V_{ab} = 5\ \Omega * 1\text{ A} = 5\text{ V}$$

Method 1:

$$V = I * (5\ \Omega || 20\ \Omega) = 1\text{ A} * (5 * 20 / (5 + 20))$$
$$= 4\text{ V}$$
$$I_L = 4\text{ V} / 20\ \Omega = 1 / 5\text{ A} = 0.2\text{ A}$$

2 steps!! -- take a shortcut by merging them: $V_{ab} = (R_{Load} * I)$, $I_L = (V_{ab} / R_{Load})$ 7

Method 2 - Voltage dividers

A very common circuit element you will see time and time again >>>

Because the current is

$$I = \frac{V_{in}}{R_1 + R_2}$$

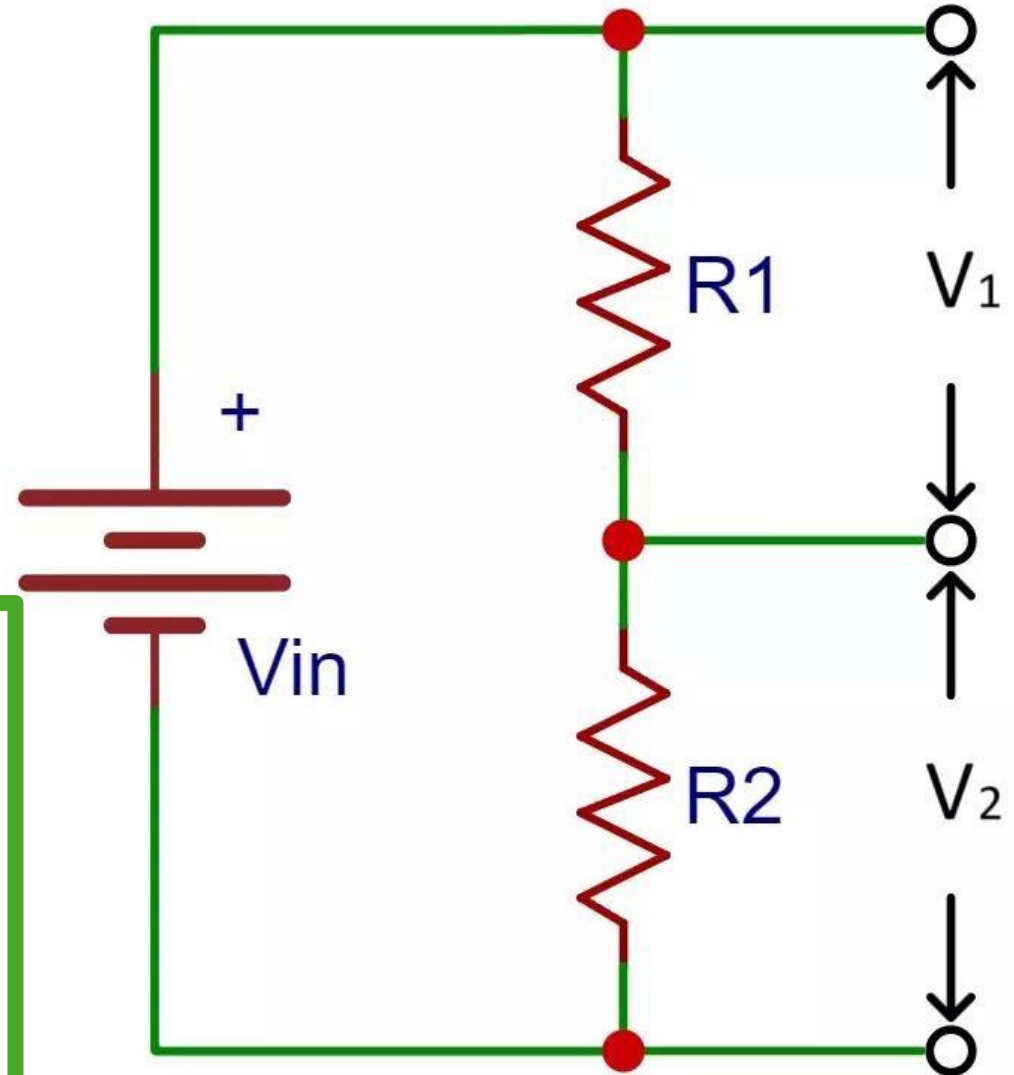
Then

$$V_1 = I * R_1 = V_{in} * \frac{R_1}{R_1 + R_2}$$

And

$$V_2 = I * R_2 = V_{in} * \frac{R_2}{R_1 + R_2}$$

Voltage divider



<<Know this by heart!

Method 2 – Current dividers

Another very common circuit element
you will see time and time again >>>

Because the voltage is

$$V_{in} = I_{in} \frac{R_1 R_2}{R_1 + R_2}$$

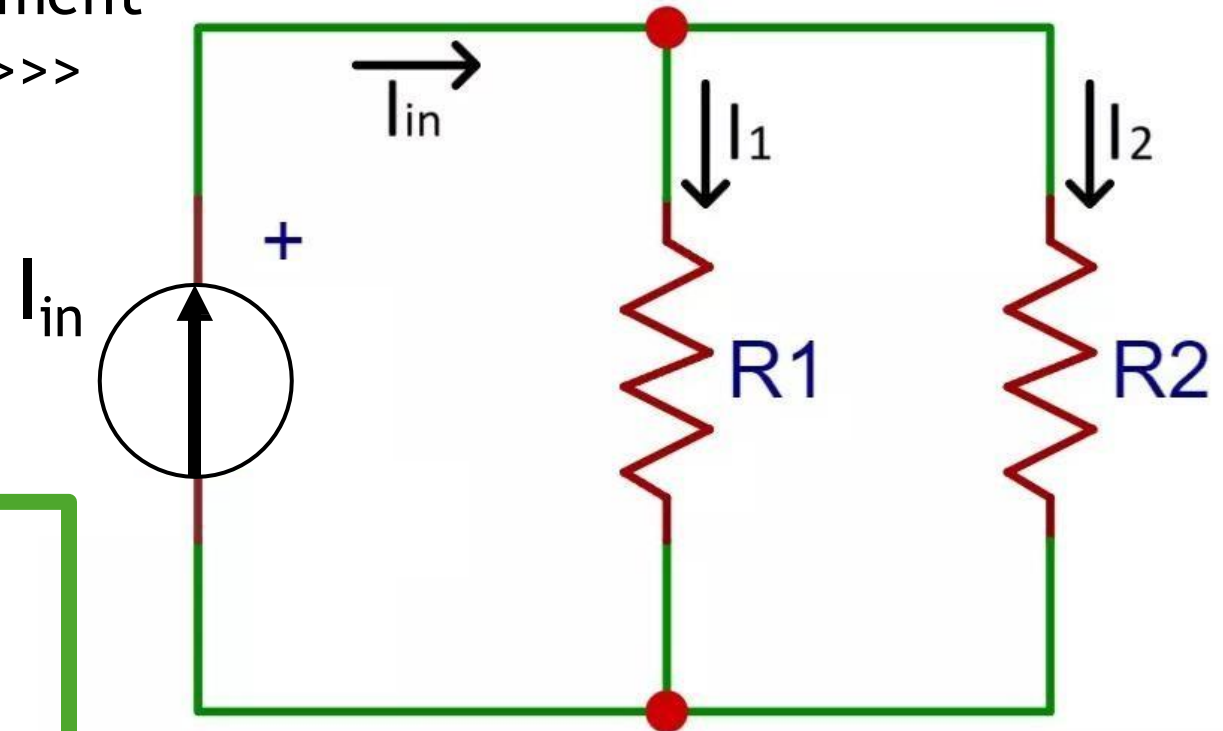
Then

$$I_1 = \frac{V_{in}}{R_1} = I_{in} * \frac{R_2}{R_1 + R_2}$$

And

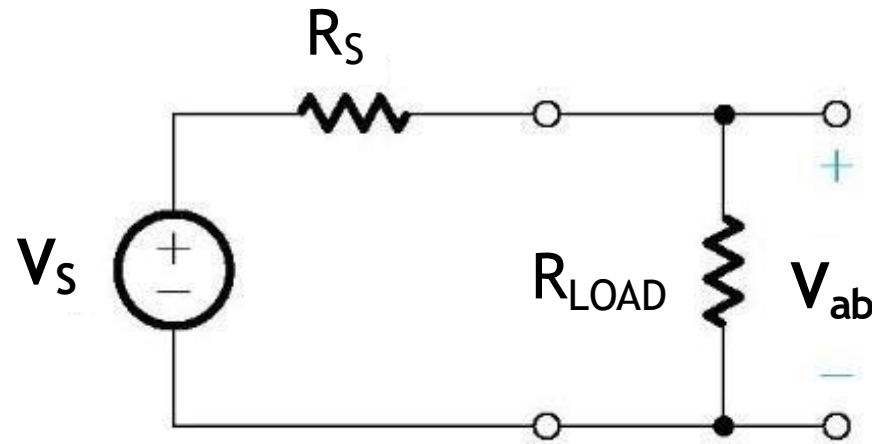
$$I_2 = \frac{V_{in}}{R_2} = I_{in} * \frac{R_1}{R_1 + R_2}$$

Current divider



<<Know this by heart!

How to make “ideal” sources



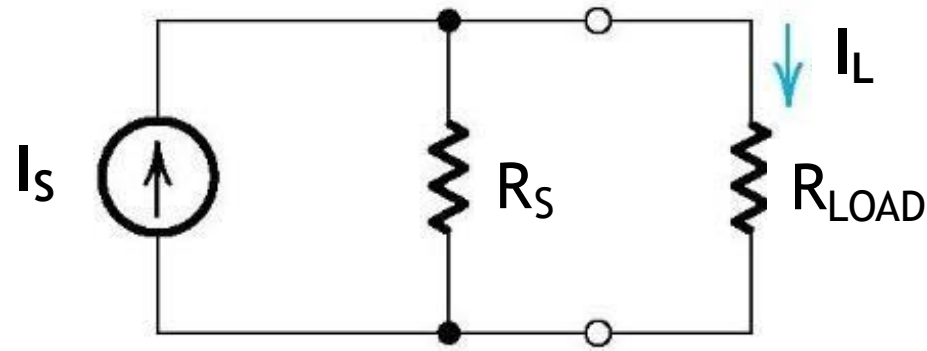
What do we need to do to make $V_{ab} = V_S$?

$$V_{ab} = \frac{R_{LOAD}}{R_{LOAD} + R_S} * V_S$$

$$1 = \frac{R_{LOAD}}{R_{LOAD} + R_S}$$

true only if $R_{LOAD} = R_{LOAD} + R_S$
so $R_S \ll R_{LOAD}$ Ideally, $R_S \rightarrow 0$

How to make “ideal” sources



What do we need to do to make $I_L = I_S$?

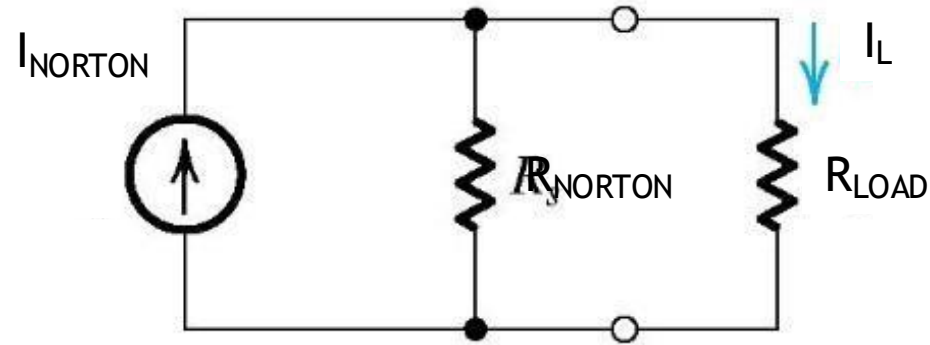
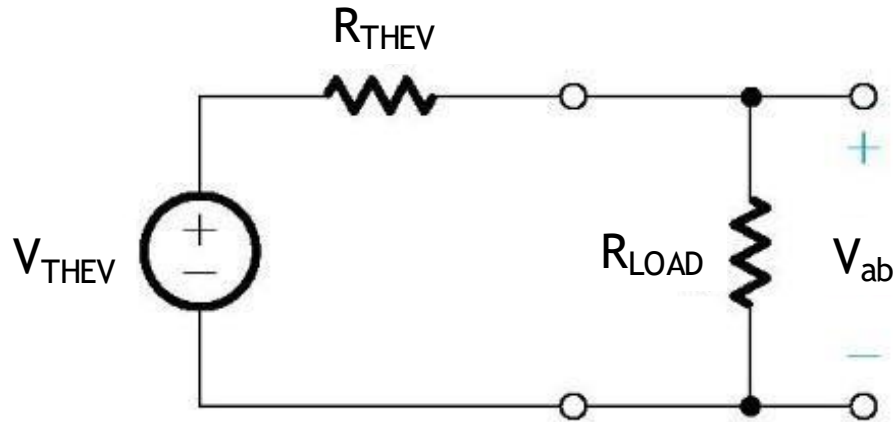
$$I_L = \frac{R_S}{R_{LOAD} + R_S} * I_S$$

$$1 = \frac{R_S}{R_{LOAD} + R_S}$$

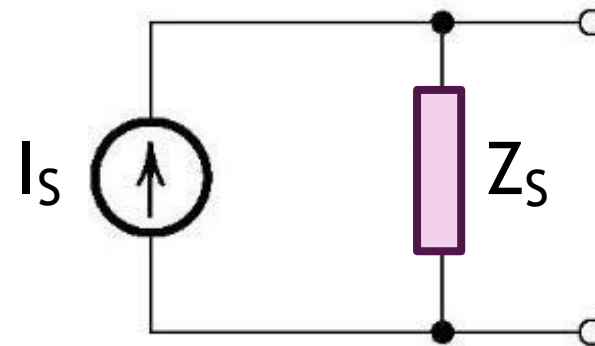
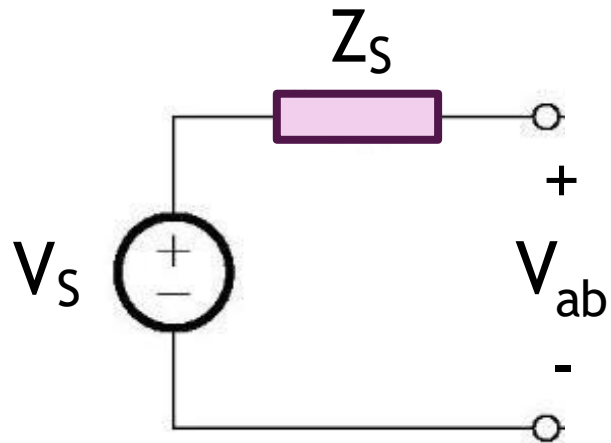
true only if $R_S = R_{LOAD} + R_S$

so $R_{LOAD} \ll R_S$ Ideally, $R_S \rightarrow \infty$

Generalizing our sources



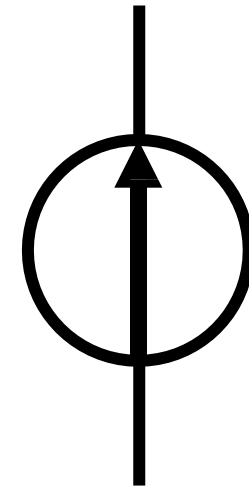
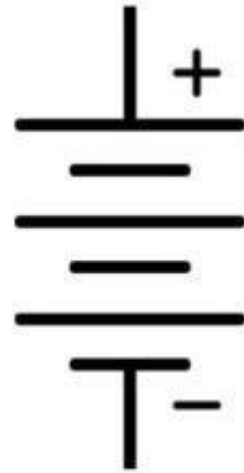
Notation up
to now



Generalized
notation (also in
your textbook)

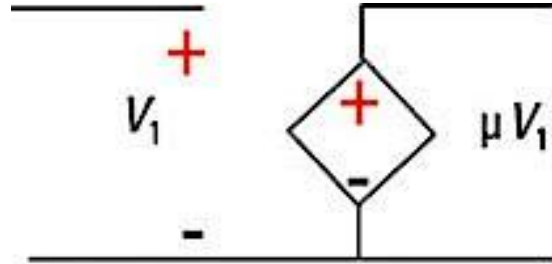
Dependent sources

Up to now, we have been dealing with independent sources

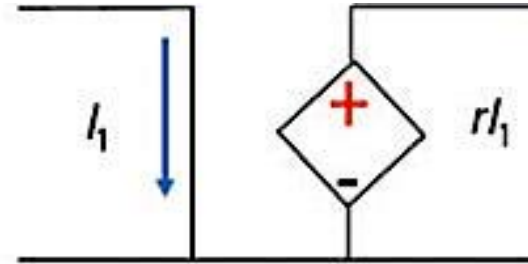


Source absorption theorem

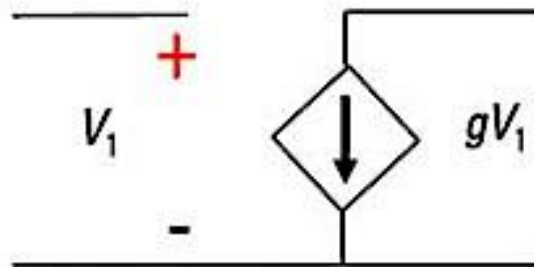
We also have dependent sources



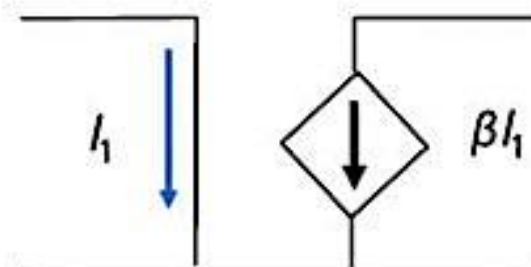
Voltage-controlled
voltage source (VCVS)



Current-controlled
voltage source (CCVS)



Voltage-controlled
current source (VCCS)



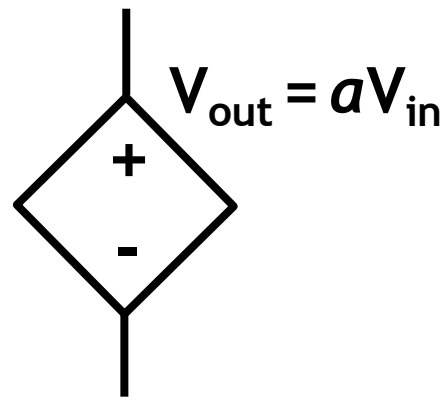
Current-controlled
current source (CCCS)

μ, r, g, β are all
scaling factors

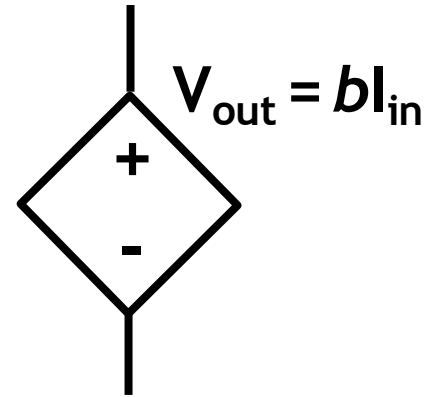
- this is where
amplification
comes from
(as we will see in
transistors)

Other dependent source forms

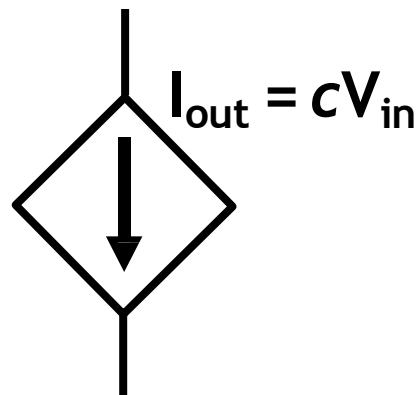
They may also show up on their own



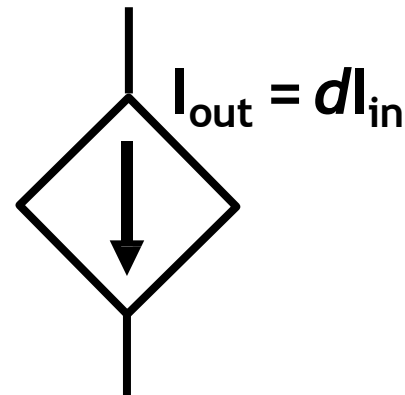
VCVS



CCVS



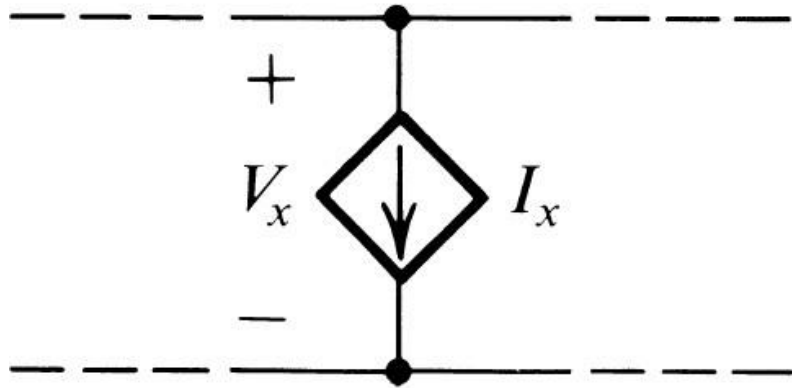
VCCS



CCCS

Source absorption theorem

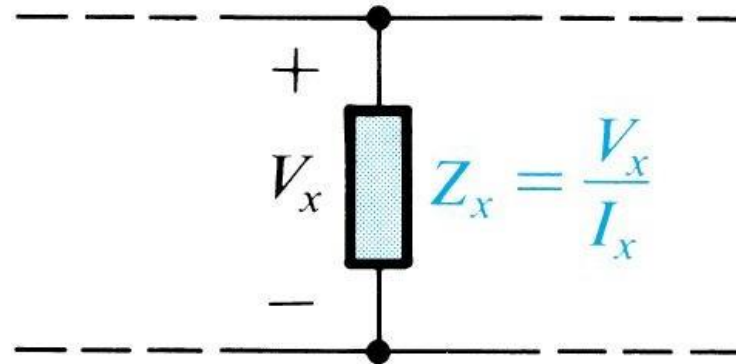
Let's say we have a current source controlled by the voltage across it



In other words, there is some conductance g_m associated with this element:

$$I_x = g_m V_x$$

So we can replace it with an impedance Z_x



Reflecting on your toolbox

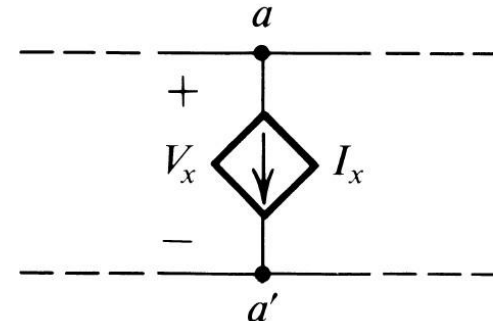
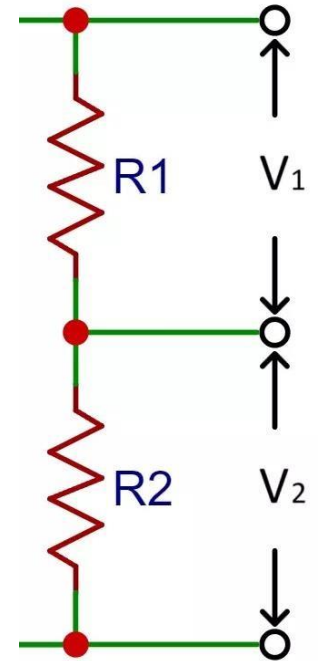
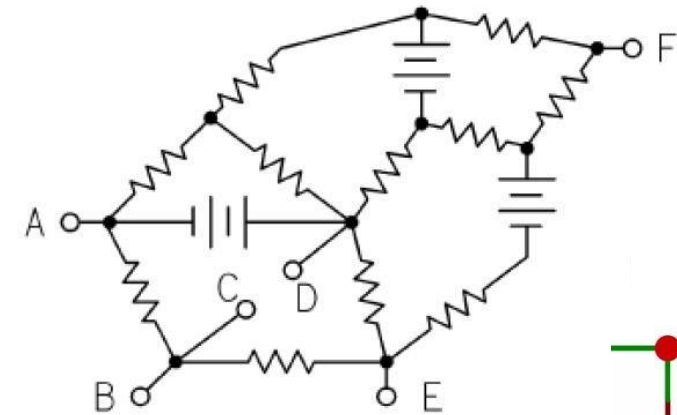
We started from transducers >>

You can now:

1. Simplify linear circuits. (Thev/Norton)
2. Determine how loading affects output voltage/current (the 2-step method).
3. Identify divider circuits (the 1-step method).
4. Simplify circuits with sources dependent on the voltage across/current through them.



Complex Linear Network



Reflecting on your toolbox

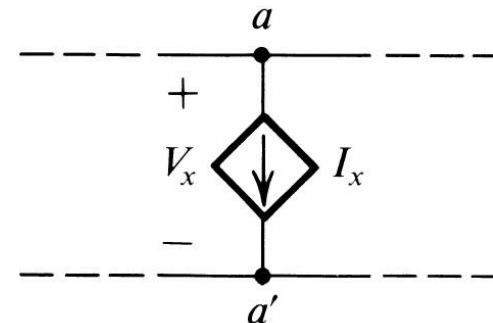
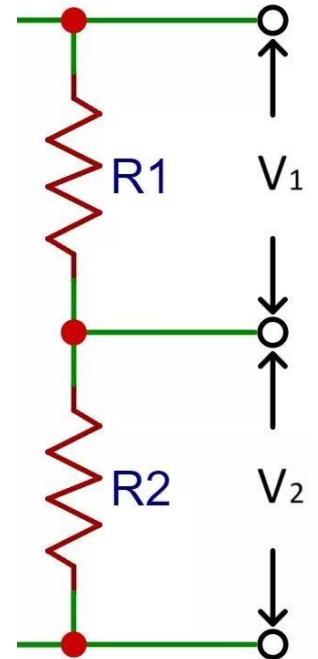
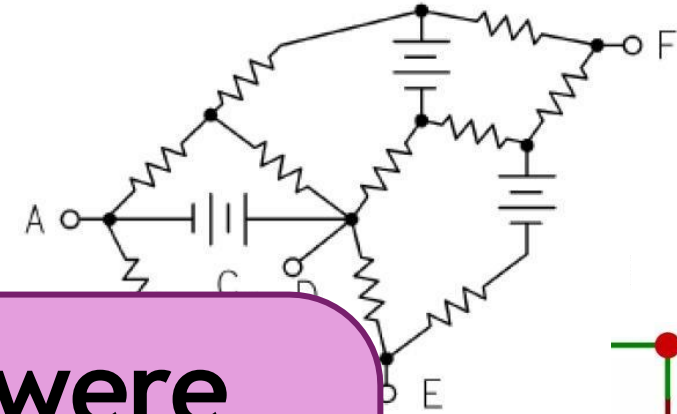
We started from transducers >>

You can now:

1. Simplify linear circuits.
2. Determine h (the 2-step n
3. Identify divider.
4. Simplify circuits with sources dependent on the voltage across/current through them.

**These analyses were
done at DC - a single
point in time!**

Complex Linear Network

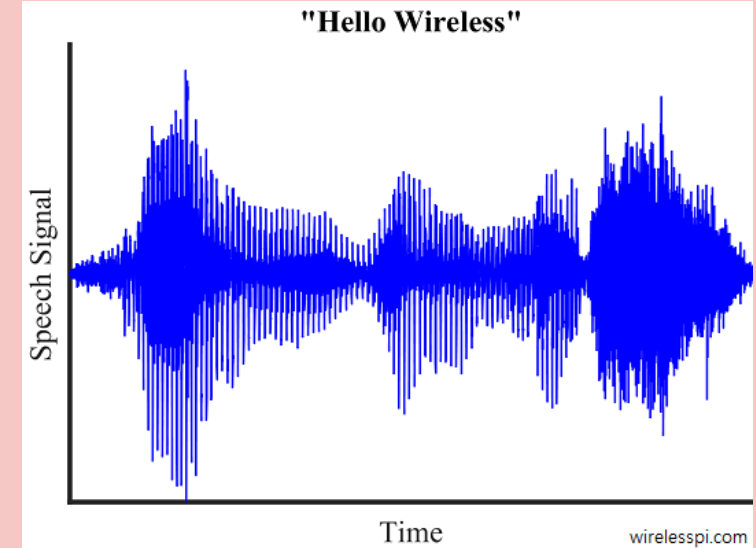


What happens over time?

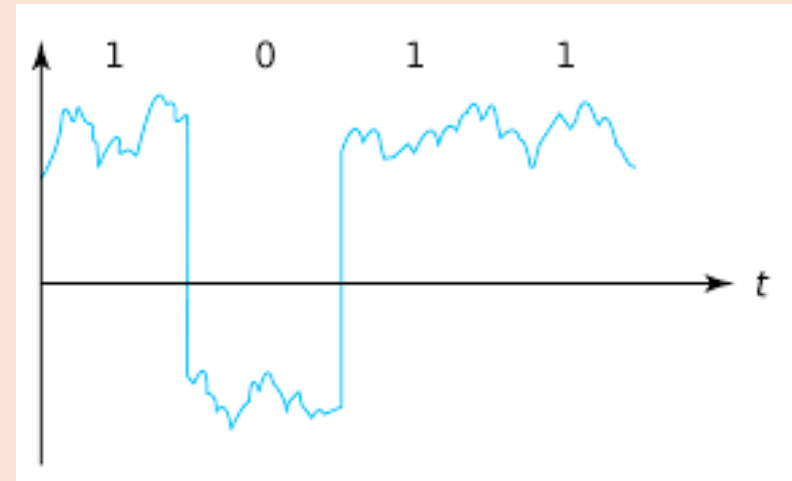
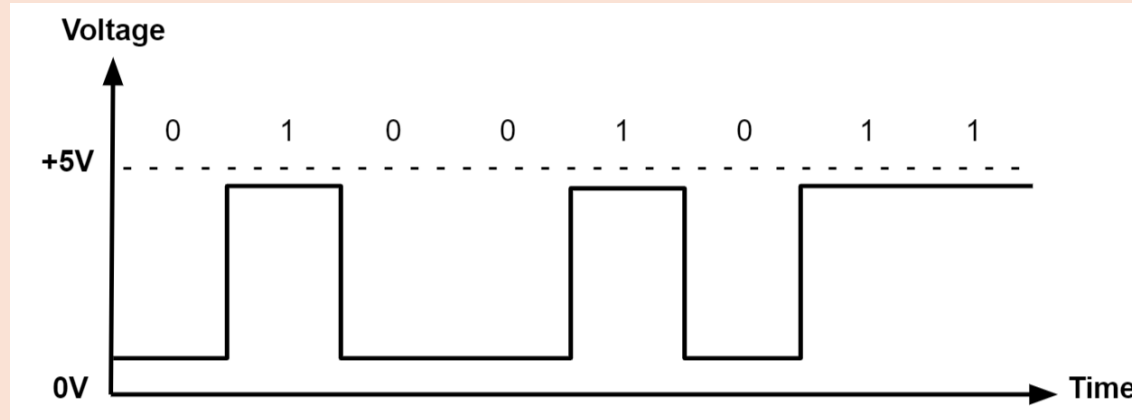
Analog



Signals change!



Digital

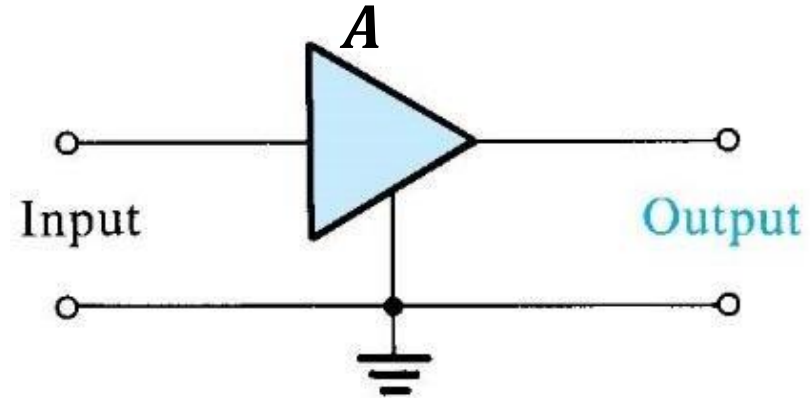
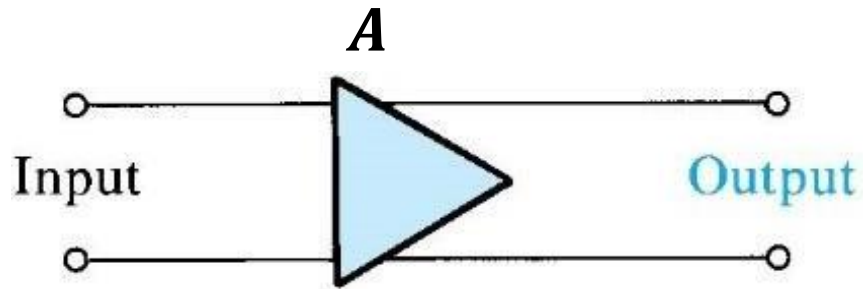


What we think it looks like

What it actually looks like

What is an amplifier?

They can look different depending on configuration



Typical rules for amplifiers:

1. Linear
2. No distortion
3. (reasonably) Low noise

This one is more common ^^

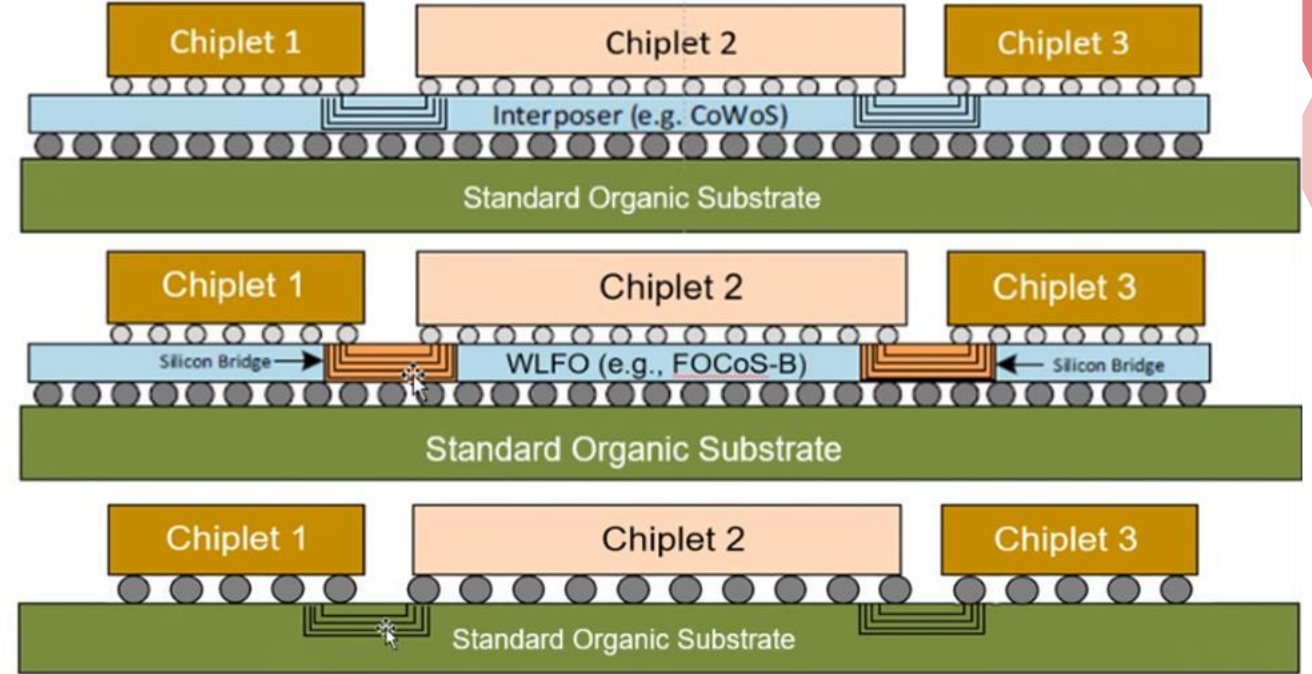
$$v_{out}(t) = A v_{in}(t)$$

Amplifier Gain

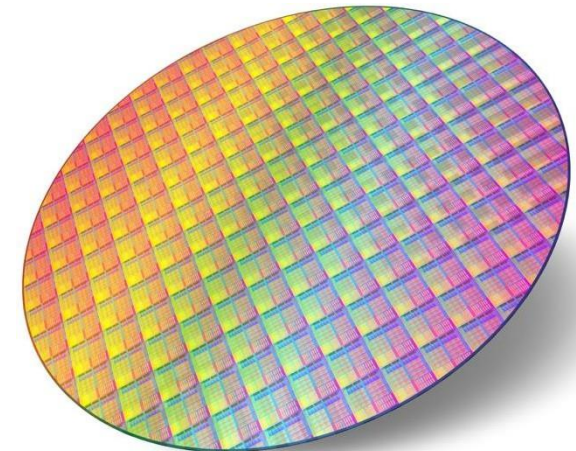
Higher gain is not always better due to saturation/clipping

Why do I want to amplify a voltage?

- Transmit my signal/power over a longer distance ($P=I^2R$ losses)
- Reliably detect small signals (e.g., from a particle detector)
- Increase my SNR (e.g., for radio receivers, Bluetooth - subject to [FCC part 15b rules](#))
- Perform switching operations (e.g., digital logic, CMOS)

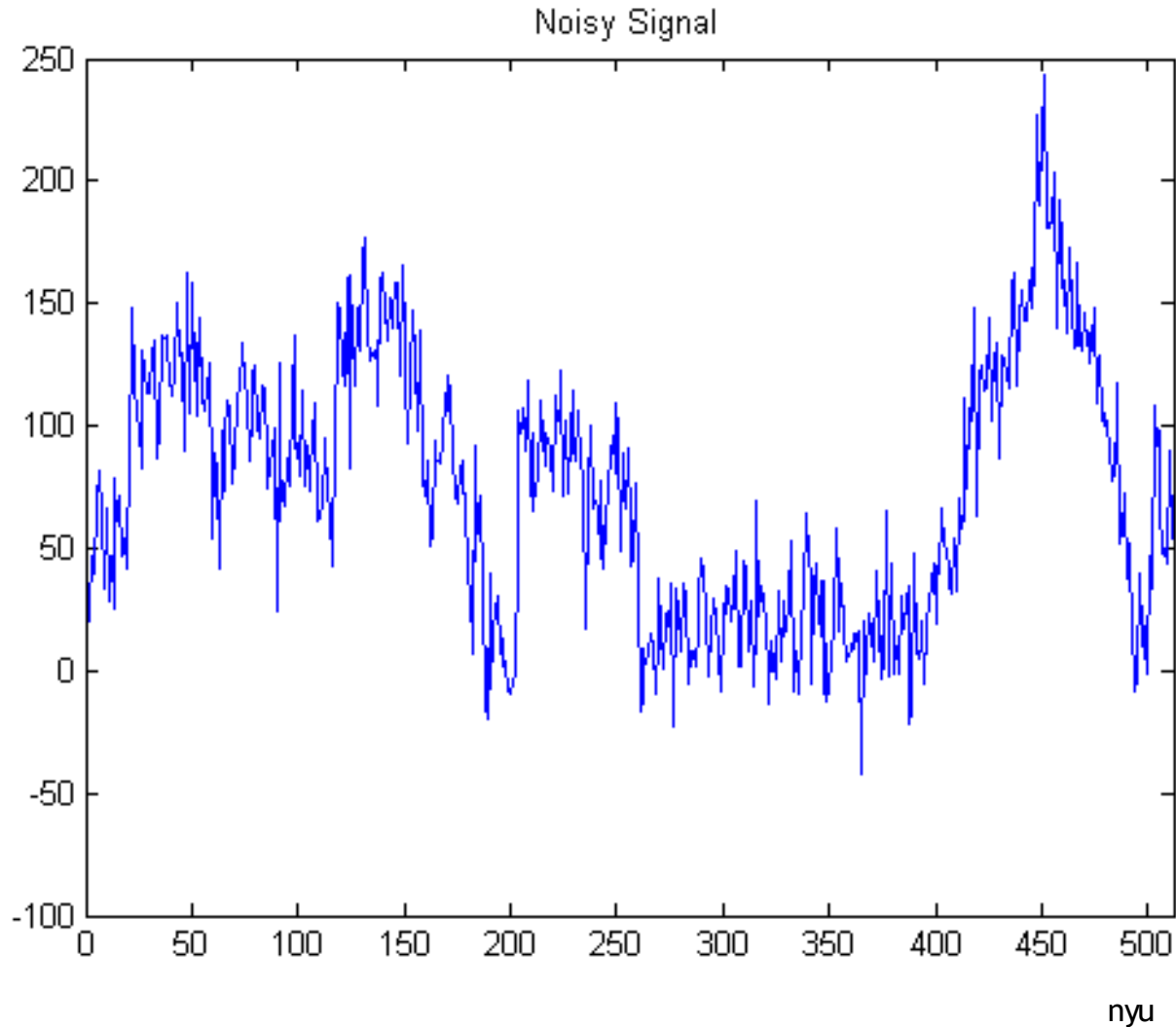


semiengineering, 2024



waferworld, 2018

Signal processing basics



Some signals can be weak & hard to process (mV, μ V, or nV regime)

So, we perform amplification!



Check out signal denoising, which is a huge topic on its own!



3 types of gain – voltage, current, power

Voltage

Helps us
know what
kind of gain
it is

$$A_v = \frac{v_{out}}{v_{in}} \text{ [V/V]}$$

$$A_v[dB] = 20 \log |A_v| \text{ [dB]}$$

(base 10)

$$|A_v| > 1 \quad \text{Gain}$$

$$A_v[dB] > 0 \quad \text{Gain}$$

$$|A_v| < 1 \quad \text{Attenuation}$$

$$A_v[dB] < 0 \quad \text{Attenuation}$$

+ 6 dB - output is 2x input
- 6 dB - output is 1/2x input

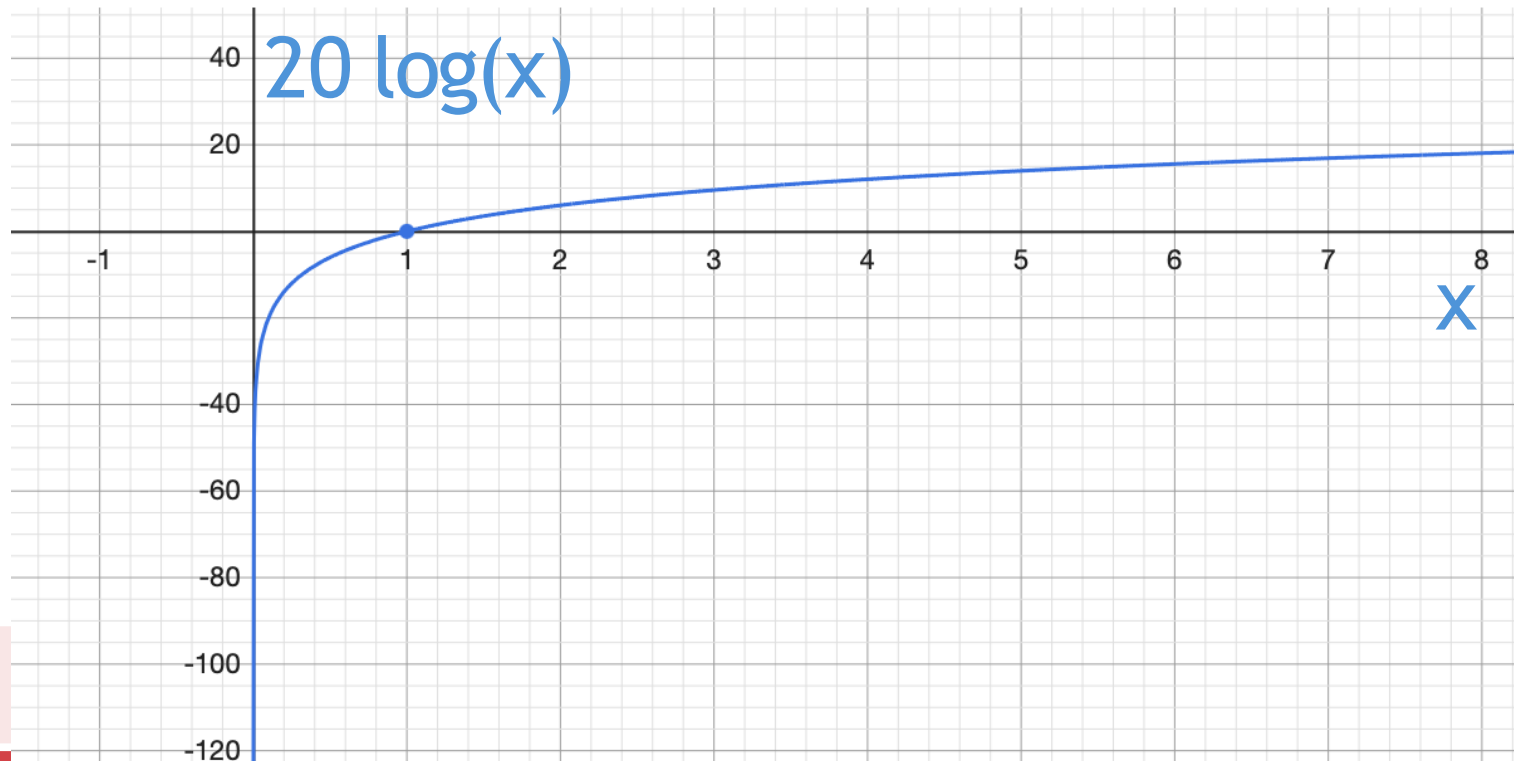
0 dB - unity gain

3 types of gain – voltage, current, power

Voltage

$$A_v = \frac{v_{out}}{v_{in}} \text{ [V/V]}$$

$$A_v[dB] = 20 \log |A_v| \text{ [dB]}$$



$A_v[dB] > 0$ Gain

$A_v[dB] < 0$ Attenuation

3 types of gain – voltage, current, power

Current

$$A_i = \frac{i_{out}}{i_{in}} \text{ [A/A]}$$

$$A_i[dB] = 20 \log |A_i| \text{ [dB]}$$

(same as voltage)

$$|A_i| > 1 \quad \text{Gain}$$

$$A_i[dB] > 0 \quad \text{Gain}$$

$$|A_i| < 1 \quad \text{Attenuation}$$

$$A_i[dB] < 0 \quad \text{Attenuation}$$

6 dB rule of thumb holds for current, too!

3 types of gain – voltage, current, power

Power

$$A_p = \frac{P_{LOAD}}{P_{in}} = \frac{v_{out} * i_{out}}{v_{in} * i_{in}} = A_v * A_i \text{ [W/W]} \quad A_p[dB] = 10 \log |A_p| \text{ [dB]}$$

$$|A_p| > 1 \quad \text{Gain}$$

$$A_p[dB] > 0 \quad \text{Gain}$$

$$|A_p| < 1 \quad \text{Attenuation}$$

$$A_p[dB] < 0 \quad \text{Attenuation}$$

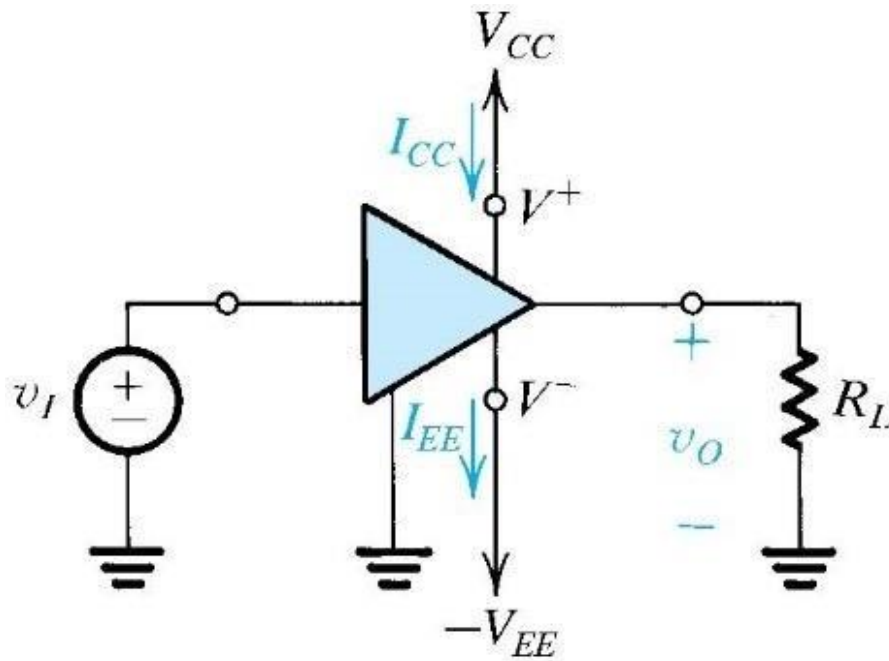
Power rule of thumb

+ 3 dB - output is 2x input

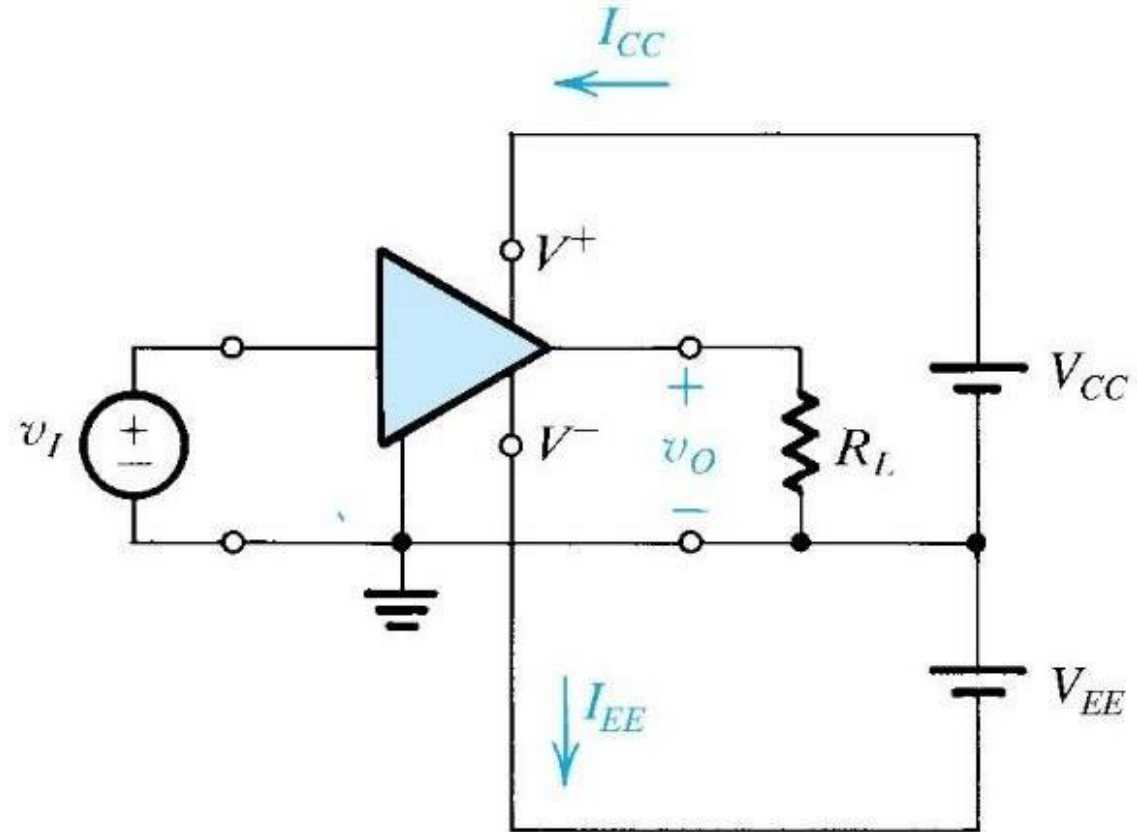
- 3 dB - output is 1/2x input

How do we connect an amp?

Step 1: Turn the power on...



Typical representation ^^

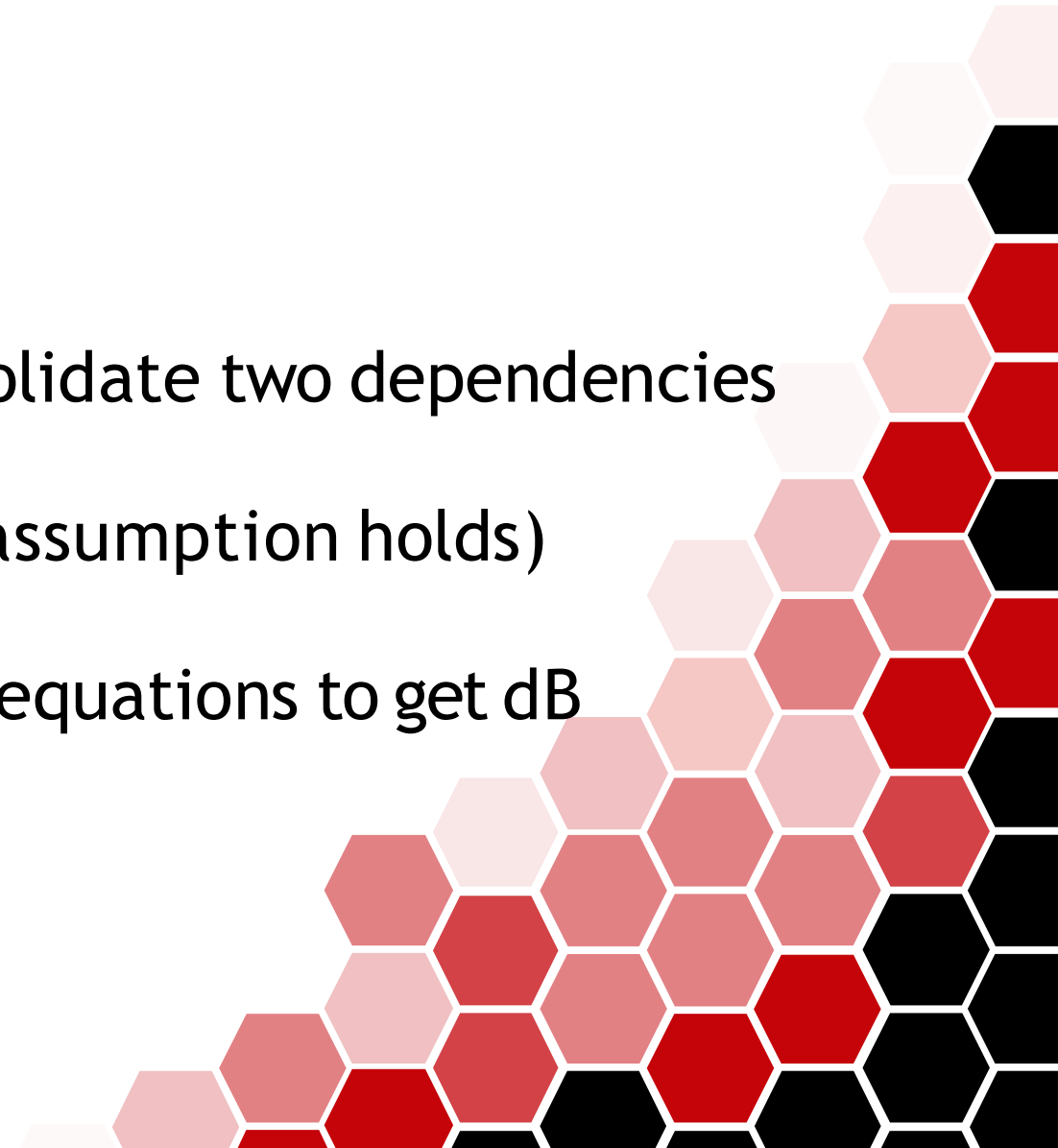


Complete representation ^^

Summary

- ✓ Recap of Lec. 1
- ✓ I- and V- dividers
- ✓ (Dependent) Source absorption - consolidate two dependencies
- ✓ Ideality of signal sources (C when this assumption holds)
- ✓ Voltage/Current/Power gain - 3 similar equations to get dB

That's all, folks!

A decorative graphic in the bottom right corner consisting of a grid of hexagons. The hexagons are colored in shades of red, pink, and black, arranged in a pattern that tapers off towards the right edge of the slide.